

SUDHARSHAN R

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SUMMARY

Second-year Electronics & Instrumentation Engineering student at KCT with hands-on experience in full-stack development (MERN/Next.js) and applied reinforcement learning. National finalist at the Meta × Scaler PyTorch OpenEnv Hackathon 2026 (selected from 31,000+ registrants), where I co-architected a production-grade RL environment with curriculum-trained LLM agents achieving a 206% score improvement on the medium evaluation tier. Seeking software engineering or ML internship opportunities to apply systems thinking to real-world problems.

EDUCATION

Kumaraguru College of Technology — Coimbatore, Tamil Nadu

Aug 2024 – May 2028

B.E. Electronics and Instrumentation Engineering | CGPA: 7.9 / 10.0

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Computer Networks, Engineering Mathematics, Embedded Systems

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java

Frameworks & Libraries: Next.js, React, Express.js, FastAPI, Tailwind CSS, Framer Motion

ML & AI: PyTorch, GRPO (RL Fine-tuning), HuggingFace Transformers, scikit-learn, Reinforcement Learning

Databases: MongoDB, MongoDB Atlas, Convex (real-time)

DevOps & Tools: Docker, Git, GitHub, Vercel, Liveblocks, Clerk, Zustand, JWT

PROJECTS

PRRegressionAuditEnv — Automated PR Regression Auditor

Jan 2026 – Feb 2026

huggingface.co/spaces/rsd-06/PRRegressionAuditEnv / github.com/rsd-06/GitPRTriageEnv

Python · FastAPI · PyTorch · GRPO · Curriculum Learning · HuggingFace · Docker | Meta × Scaler OpenEnv Hackathon 2026

- Architected a **full OpenEnv-spec RL environment** from scratch where **LLM agents act as automated senior code reviewers**, detecting accidental regressions in GitHub PRs across 3 difficulty tiers (Easy / Medium / Hard).
- Engineered a **deterministic RLVR reward system** with zero LLM judges, eliminating persuasion-based reward hacking; implemented the **GuardSuite** — 4 anti-reward-hacking modules (KeywordStuffingDetector, RepetitionDetector, FixQualityValidator, TimingGuard).
- Fine-tuned **Qwen2.5-1.5B-Instruct** via **GRPO with Curriculum Learning** (Bootstrap → Intermediate → Advanced phases), achieving **91% improvement (Easy), 206% (Medium), 130% (Hard)** over the untrained baseline across 45 evaluation episodes.
- Deployed a **4-agent specialist pipeline** (SafetyGate → DefectLocator → ReviewerRouter → ReviewComment) as a **Dockerized FastAPI service on HuggingFace Spaces** with live W&B dashboards, a curriculum state API, and a guard audit log.

SkillSync — Skill-Based Collaboration Platform

Sep 2024 – Present

https://skill-sync-three-tan.vercel.app/login / github.com/rsd-06/SkillSync

Next.js · Express.js · MongoDB Atlas · Tailwind CSS · Zustand · JWT · Docker · REST API

- Developed a **full-stack platform** for engineering-student project collaboration, designing a **skill-graph matching engine** that pairs users by complementary skill gaps — replacing fragmented WhatsApp-group coordination.

- Implemented **JWT authentication with refresh tokens**, Agile-style Kanban team workspaces, and a **contribution-based reputation scoring system**; onboarded 8 beta users and currently being evaluated for **KCT Incubator funding**.
- Containerized with **Docker**; deployed frontend on **Vercel** and REST API on **Render** with indexed **MongoDB Atlas** queries optimized for skill-based lookup across user profiles.

Google Docs Clone — Real-Time Collaborative Editor

Jul 2024 – Aug 2024

<https://skill-sync-three-tan.vercel.app/login> / github.com/rsd-06/googleDocsMini

Next.js · TypeScript · Liveblocks · Convex · Clerk · Vercel | Live: google-docs-mini.vercel.app

- Engineered a **full-stack real-time collaborative document editor** with concurrent multi-user editing via **Liveblocks** (CRDT-based sync) and **Convex** real-time backend, ensuring zero-conflict shared document state.
- Integrated **Clerk** for secure OAuth and built a complete **rich-text toolbar** (bold, italic, headings, lists, alignment) with persistent state — deployed to production on **Vercel** using full **TypeScript** for type-safe API contracts.

EXPERIENCE & ACTIVITIES

Open Source Contributor — Oppia Foundation — github.com/oppia

2025 – Present

- Onboarding as a contributor to **Oppia** (10,000+ GitHub stars) — an open-source adaptive learning platform; studying codebase architecture, triaging issues, and working toward first PR submission.

Model United Nations — Security Council Delegate — Kumaraguru College of Technology

2024

- Represented **Algeria** on the Security Council; delivered structured geopolitical speeches and led clause-level negotiation — awarded **Verbal Mention** for argumentation quality.

Dengue Outbreak Prediction — AI Research Project — KCT, under Dr. Jeya Daisy

2024

- Co-developed a district-level dengue risk prediction platform using **Random Forest** (87% holdout accuracy) with climate and epidemiological features; built **Flask REST API + Leaflet.js** interactive hotspot map for health-official deployment — formal academic report published.

AWARDS & ACHIEVEMENTS

- **National Finalist — Meta × Scaler PyTorch OpenEnv Hackathon 2026:** Selected from 31,000+ registrants; submitted a production-grade RL environment with trained multi-agent code-reviewer system.
- **3× Ideathon Winner — Kumaraguru College of Technology:** Won three consecutive internal ideathon rounds for problem formulation and technical solution pitching.
- **MUN Verbal Mention — Security Council, KCT MUN 2024:** Recognized for structured argumentation and policy research representing Algeria.